MILITARY SPECIFICATION

OSCILLATOR SUBASSEMBLY (NAVORD Dwg 1563736) (FOR TORPEDO MARK 37 MODS 0, 2, and 3)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification establishes the requirements for the procurement of an Oscillator Subassembly, NAVORD Drawing 1563736, which provides the frequency for the oscillator and amplifier assembly,

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein

SPECIFICATIONS

Federal

PPP-B-621	Box, Wood, Nailed and Lock-corner

Military

MIL-E-1	Electron Tube; General Specification for
MIL-P-116	Preservation, Methods of
MIL-E-82590	Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

FSC 1355

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:01Z
Check the source to verify that this is the current version before use.

MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-type Base
MIL-STD-200	Electron Tube and Semi-conductor Devices, Diodes
MIL-STD-202	Test Methods for Electronic and Electrical Parts

DRAWINGS

Naval Ordnance Systems Command

NAVORD Dwg 1563736	Oscillator Subassembly

(Copies of specifications, standards, drawings, and publications requested by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATION

Handbook H28	Screw Thread Standards for Federal Service

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

UNITED STATES OF AMERICA STANDARDS INSTITUTE

USAS Y32.2 1967	Graphic Symbols for Electrical and Electronics Diagrams

(Applications for copies should be addressed to DOD Single Stock Point, U. S. Naval Supply Depot, Philadelphia, Pennsylvania 19120.)

3. REQUIREMENTS

3.1 Description. The oscillator subassembly is a plug-in electrical circuit consisting of a center-tapped coil and a capacitor which is sealed in a metal container. These components are connected to form the tuned circuit portion of a Hartley oscillator for the oscillator and amplifier assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:01Z
Check the source to verify that this is the current version before use.

3.2 General Requirements

3.2.1 Data required. No data is required by this specification (other than reports accompanying samples submitted for preproduction and periodic production testing), or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3,2.2 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the oscillator subassembly shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing oscillator subassemblies that comply with the technical requirements of the contractor. See 4.3.1.

3.2.3 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the oscillator shall conform strictly to the detailed requirements indicated on the applicable drawings, specifications, and publications listed in NAVORD Dwg 1563736.

3.2.4 Surface treatment, construction and dimensions. Surface treatment, construction and dimensions for the oscillator shall be in accordance with the applicable drawings, specifications, and publications listed in NAVORD Dwg 1563736.

3.2.5 Moisture, fungus, and corrosion. The oscillator shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified on the drawings and specifications listed in NAVORD Dwg 1563736,

3.2.6 Marking. Marking of assemblies. subassemblies, and parts shall be in accordance with applicable drawings listed in NAVORD Dwg 1553736.

3.2.7 Threads. Unless otherwise specified: all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.3 Symbols. Interpret schematic diagram symbols in accordance with USAS Y32.2.

3.4 Performance requirements and product characteristics, The oscillator subassembly shall meet the following performance requirements and product characteristics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:01Z
Check the source to verify that this is the current version before use.

3.4. 1 Frequency The frequency of the oscillator shall be 1000 plus or minus 3 Hz.

3.4.2 Voltage. The voltage at pin 7 of the oscillator subassembly shall be 34 plus or minus 2 vrms when measured as specified in (4.4.4.7).

3.4.3 Voltage balance. The voltage at pin 1 of the oscillator subassembly shall be within plus or minus 0.5 vrms of the voltage measured in 3.4.2, when measured as specified in 4.4.4.3.

3.4.4 Waveform. The output voltage of the oscillator shall be sinusoidal, free from noise, spikes, or other transients, The harmonic distortion shall not exceed two percent (%). The distortion factor, D, is expressed as follows:

$$D\% = 100 \sqrt{\frac{\text{Sum of the squares of amplitudes of harmonics}}{\text{Square of amplitude of fundamental}}}$$

3.4.5 Phase difference. There shall be a 180 plus or minus 2 degrees (0) phase difference of the voltages at pin 1 and pin 7 of the oscillator subassembly.

3.4.6 Dielectric strength. There shall be no evidence of breakdown when 500 vrms at 60 Hz is applied for one minute between mutually insulated terminal and case.

3.4.7 Insulation resistance. The insulation resistance shall be 10,000 megohms minimum between mutually insulated terminals and case when measured at a test potential of 100 vdc. The insulation resistance measurement shall be made immediately after the dielectric strength potential is removed in performing the test to meet the specified requirement in 3.4.6.

3.4.8 Operating life. The oscillator subassembly operating life shall be at least 800 cycles of operation when tested as specified in 4.5.5.

3.5 Environmental requirements. The oscillator subassembly shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.5.

3.5.1 Vibration, The oscillator subassembly shall meet the following vibration requirements:

3.5.1.1 Vibration, nonoperating. The oscillator subassembly shall show no damage and shall meet all the performance requirements and product

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:01Z
Check the source to verify that this is the current version before use.

characteristics of this specification after being tested as in 4.5. 1.2.

3.5.1.2 Vibration, operating. The oscillator subassembly shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.5.1.3.

3.5.1.3 Vibration, transportation. The oscillator subassembly shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.1.4.

3.5.2 Shock, transportation. The oscillator subassembly shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.2.

3.5.3 Temperature and humidity

3.5.3.1 High temperature and humidity, nonoperating. The oscillator subassembly shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.3.1.

3.5.3.2 High temperature, operating. The oscillator subassembly shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.3.2.

3.5.3.3 Low temperature, nonoperating. The oscillator subassembly shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.3.3.

3.5.3.4 Low temperature, operating. The oscillator subassembly shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.3.4.

3.5.4 Immersion. The oscillator shall show no evidence of leakage during the immersion test, specified in 4.5.4, and thereafter shall meet the performance and product requirements of this specification,

3.6 Cleanup Prior to and after final assembly all parts, components and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips and other foreign matter. Burrs and sharp edges as well as excessive rosin flash shall be removed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:01Z
Check the source to verify that this is the current version before use.

3.7 Workmanship. The oscillator, including all parts and accessories, shall be constructed and finished in a reamer to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges,

3.7.1 The standards of workmanship exhibited in any approved pre-production sample, subject to any qualification stated in the Governments notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

k. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where any of the inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the oscillator sub-assembly shall be classified as follows:

(a) Preproduction inspection (paragraph 4.5)

(b) Quality conformance inspection (paragraph 4.4)

(c) Periodic production inspection (paragraph 4.5)

4.3 Sampling

4.3.1 Reproduction, A preproduction sample of five oscillators, manufactured in accordance with 3.2.2, shall be subjected to the pre-production tests detailed in 4.5 at an activity designated by the pro-curing actitity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the oscillator by the supplier.

Source: https://assist.dla.mil -- Downloaded: 202640-23T05:01Z
Check the source to verify that this is the current version before use.

prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Lot size, Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.4, a random sample of five oscillators will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The oscillators shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

4.4 Quality conformance inspections. Each oscillator delivered under contract shall be su-ejected to the following tests and examination. Failure to satisfactorily pass any test specified herein shall be considered cause for rejection.

4.4.1 Test conditions. Unless otherwise specified, the oscillator subassembly shall be subjected to quality conformance tests under the following conditions:

4.4.1.1 Temperature. Room ambient 65 degrees F to 95 degrees F (18.5 degrees C to 35 degrees C).

4.4.1.2 Altitude. Normal ground.

4.4.1.3 Vibration. None.

4.4.1.4 Humidity. Room ambient to 95 percent relative maximum.

4,4.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment, facilities and provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity, and when definite requirements are not specified they shall be of sufficient accuracy and quality to permit performance of the required quality conformance tests.

4.4,3 Test Equipment. The following items of test equipment are required

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:01Z
Check the source to verify that this is the current version before use.

to perform the quality conformance tests set forth in this specification.

4,4.3.1 Voltmeter. Ballantine Model 303 or Model 300G, or equal,

4.4.3.2 Counting, timing and frequency measuring meter. Universal EPUT and Timer, Model 7360, or equal.

4.4.3.3 Cathode-ray oscilloscope. OS-48/U.

4.4.3.4 Wave analyzer. TS-615/U.

4.4.3.5 Insulation tester. Radio Frequency Laboratories 184c.

4.4.3.6 Dielectric tester. MOD NMD/Y.

4.4,3.7 Power supply. A power supply that will furnish the power specified as follows. The power supply shall have filament power and plate power switches in the output circuits of the test circuit (see Figure 1).

4.4.3.7.1 Heater power. The electrical power which is used to operate the 6C4WA in the oscillator test circuit shall meet the following requirements. All values shall be measured at the power supply output terminals.

(a) No-load voltage. With no load the voltage shall not exceed minus 7.5 volts direct current (vdc).

(b) Current. The heater will require a current of 0.15 ampere.

(c) Voltage. Under a load current of 0.15 ampere, the voltage shall be minus 6.25 plus or minus 0.10 vdc.

(d) Internal impedance. The impedance which the power source presents to the heater at a load current of 0.15 ampere shall not exceed 50 ohms at 1 KHz,

(e) Alternating-current (ac) voltage components. Under a load current of 0.15 ampere, the ac voltage components shall not exceed the following:

(1) 30 Hertz (Hz) to 150 kilohertz (KHz) ----0.09 volt root-mean-square (vrms).

(2) 0.9 KHz to 1.1 KHz----9O microvolt rms

4.4.3.7.2 Plate power. The electrical power required to operate the 6C4WA plate in the oscillator subassembly test circuit shall meet the following requirements. All values specified shall be measured at the power supply output terminals.

(a) No-load voltage. With no load the voltage shall not exceed

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:01Z
Check the source to verify that this is the current version before use.

165 vdc.

(b) Current. The test circuit shall require a current of 6 milli-amperes (ma).

(c) Voltage, Under a load current of 6 ma, the voltage shall be 150 plus or minus 1 vdc.

(d) Internal impedance. The impedance which the power source presents to the test circuit at a load current of 6 ma shall not exceed 5 ohms at 1 KHz.

(e) AC Voltage components. Under a load current of 6 ma, the ac voltage components shall not exceed the following:

(1) 30 Hz to 150 KHZ-------0.2 vrms

(2) 0.9 KHz to 1.1 KH2-----2. O millivolts rms

4,4.3.8 Power source. The source shall be 115 vrms, 60 Hz, single phase.

4.4.3.9 Inductor. 4.3 henries plus or minus 1 percent dc resistance 430 ohms plus or minus 5 -percent, Q of 65 minimum and 95 maximum at 4000 Hz plus or minus 5 percent, distributed capacitance 100 ndcromicrofarad plus or minus 20 percent.

4.4.3.10 Capacitor. 0.47 microfarad plus or minus 5 percent, 200 V.

4.4.3.11 Capacitor. 0.15 microfarad plus or minus 5 percent, 200 V.

4.4.3.12 Capacitor, 4700 picofarad plus or minus 1 percent, 500 V; 2 required.

4.4.3.13 Resistor. 2.37 K plus or minus 1/2 percent, 1/2 W.

4.4.3.14 Resistor. 38,6 K plus or minus 1/2 percent, 1/2 W: 2 required.

4.4,3.15 Resistor. 100,000 ohms plus or minus 1 percent, 1/2 W; 2 required.

4.4.3.16 Resistor. 120,000 ohms plus or minus 1 percent, 1/2 W; 2 required.

4,4.3.17 Resistor. 1800 ohms plus or minus 1 percent, 1/2 W.

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:01Z
Check the source to verify that this is the current version before use.

4.4.3.18 Electron tube. 6C4WA; MIL-STD-200 designation JAN6C4WA. The contractor shall test this tube in accordance with the procedures of MIL-E-1. This tube shall meet the requirements of MIL-E-1 except that the requirement for gain shall be 95 -volts peak-to-peak out relative to 14 volts peak-to-peak applied to the grid when tested in circuit of Figure 1 under the following conditions:

(a) Oscillator subassembly disconnected.

(b) Capacitor (4.4.3.12) disconnected at junction of capacitors (4.4.3.11 and 4.4.3.12).

(c) Resistor (4.4.3.14) disconnected at junction of capacitor (4.4.3.11) and resistor (4.4.3.14).

(d) Output voltage measured at depicted point 7 of Figure 1.

4.4,3.19 Test leads, connectors, sockets, and small tools as required.

4.4.4 Test Procedure. The following tests constitute the quality conformance tests of the oscillator subassembly

4,4.4.1 Orientation. The oscillator subassembly, with respect to the 6C4WA electron tube, shall be oriented so that terminal one of the oscillator subassembly shall ‘be minimum distance from terminal six of the electron tube.

4.4.4.2 Unless otherwise specified, all quality conformance tests shall be conducted using the test equipment in 4.4.3 and the test circuits shall be connected as described herein.

4.4.4.3 Connect the voltmeter (4.4.3.1), electronic counter (4.4.3.2), oscilloscope (4.4.3,3), wave analyzer (4.4.3.4), insulation tester (4. 4.3.5), dielectric tester (4.4.3.6), and power supply (4.4.3.7) to the power source (4.4,3.8). Set the power stitches of these instruments to ON, and allow them to warm up for the time necessary for the them to stabilize.

4,4.4.4 To measure the dielectric withstanding voltage as set forth in 3.4.6:

4.4.4.4.1 Connect the dielectric tester to pin 3 and the mounting case (mounting stud) and apply 500 vrms, 60 HZ, for one minute. There shall be no evidence of breakdown.

4.4.4.5 To measure the insulation resistance as set forth in 3.4.7:

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:01Z
Check the source to verify that this is the current version before use.

4.4.4.5.1 Immediately after completing 4,4.4.4.1, remove the dielectric tester leads and connect the insulation tester leads to pin 3 and the case of the oscillator subassembly. Measure the resistance at a test potential of 100 vdc. The resistance shall be 10,000 megohms minimum Remove the insulation tester leads.

4.4.4.6 To measure the frequency as set forth in 3.4.1:

4.4.4.6.1 Set the power supply switches to OFF. Connect the oscillator test circuit as shown in Figure 1. Careful placement of the leads may be necessary to prevent pickup

4.4.4.6.2 In accordance with the instructions provided with the electronic counter, set the controls to measure frequency. Connect the counter to pin 1 of the oscillator. All measurements shall be made with reference to ground. The test instrument leads shall be connected to pin 3 of the oscillator subassembly, which shall be connected to ground.

4.4.4.6.3 Set heater and plate power switches to ON. Allow the test circuit to warm-up for the time necessary for it to stabalize.

4.4.4.6.4 Record the frequency shown on the counter. This frequency shall meet the requirement of 3.4.1.

4.4.4.6,5 Remove the counter lead from pin 1.

4.4,4.7 To measure the voltage as set forth in 3.4.2:

4.4.4.7.1 Connect the voltmeter (4.4,3.1) lead to pin 7 of the oscillator subassembly.

4.4,4.7.2 Record the voltage indicated on the meter. The voltage shall meet the requirement of 3.4.2.

4.4.4.8 To measure the voltage balance as set forth in 3.4.3:

4.4,4.8.1 Remove the voltmeter lead from pin 7. Connect this lead to pin 1 of the oscillator subassembly,

4.4,4.8.2 Read and record the voltage indicated on the voltmeter.

4.4.4.8.3 The voltage shall be within 0.5 vrms of the voltage recorded in 4.4.4.7.2.

4.4.4.9 To measure the waveforms set forth in 3.4.4:

4.4.4.9.1 Remove the voltmeter lead from pin 1 of the oscillator subassembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:01Z
Check the source to verify that this is the current version before use.

4.4.4.9.2 Connect the oscilloscope (4.4.3.3) to pin 1 of the oscillator subassembly,

4.4.4.9.3 Observe the waveform on the oscilloscope. The wave shall be sinusoidal and shall not contain noise or spikes (transients).

4.4.4.9,4 Remove the oscilloscope lead from pin 1 and connect the wave analyzer (4.4.3.4) to pin 1 of the oscillator subassembly.

4.4.4.9.5 In accordance with the instructions provided with the wave analyzer, measure and record the amplitude of the fundamental and the amplitudes of es many harmonics as possible.

4.4.4.9.6 Using the values recorded in 4.4.4.9,5 and the formula given in 3.4.4 compute the distortion factor “D”, The distortion factor “D” shall not exceed two percent.

4.4.4.9.7 Remove the wave analyzer lead from pin 1 of the oscillator subassembly.

4.4.4.10 To measure the phase difference as set forth in 3.4.5:

4.4.4.10.1 In accordance with the instructions provided with the Universal EPUT meter (4.4.3.2), set the meter controls for Time-Interval-Meter (TIM) Operation.

4.4.4.10.2 Connect pin 1 of the oscillator subassembly to the B input channel.

4.4,4.10.3 Connect pin 7 of the oscillator subassembly to the A input channel,

4.4.4.10.4 With the TIME-SECONDS selector on 10^{-6}, the ATTEN. selector on 10 ac, both TRIGGER VOLTAGE controls to their center positions and the SLOPE switches on plus (+), read the count on the EPUT meter,

4.4.4.10.5 Set the SLOPE switches to minus (-), and read the count. If this reading is not the same es 4.4.4.10.4, adjust the TRIGGER VOLTAGE controls until the count is the same for both plus and minus SLOPE switch settings.

4.4.4.10.6 Record the count.

4.4.4.10.7 Compute the phase angle from the following formula:

$$\phi = F \times t \times 360 \text{ degrees}$$

where,

ϕ = the phase angle in degrees,

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:01Z
Check the source to verify that this is the current version before use.

t = count recorded in 4.4.4.10.5 in seconds (usec x 10^{-6})

f = frequency recorded in 4.4.4.6.4 in Hz.

The phase shall meet the requirement set forth in 3.4.5.

4.4.4.10.8 Remove the EPUT leads from pins 1 and 7 of the oscillator subassembly, remove all power.

4.4.4.11 Visual and dimensional examination. Each oscillator shall be examined to deter-mine conformance with the requirements for workmanship and dimensions.

4.4.4.12 Packaging, packing, and marking. Packaging, packing, and marking shall comply with the requirements of Section 5.

4.5 Reproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.4, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to t-he procuring activity.

4.5.1 Vibration

4.5.1.1 Mounting. The oscillator assembly, less pack or packaging, shall be mounted to a test fixture as shown in Figure 2 and tested as specified in 4.5.1.2 and 4.5.1.3.

4.5.1.2 Vibration, nonoperating. The oscillator subassembly shall be subjected to the "Vibration, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.1.1.

4.5.1.3 Vibration. operating. The oscillator subassembly shall be subjected to t-he "Vibration, operating" test of MIL-E-82590 and during and thereafter shall meet the requirements of 3.5.1.2.

4.5.1.4 Vibration, transportation. The oscillator subassembly shall be subjected to the "Vibration, transportation" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5,1.3.

4.5.2 Shock, transportation. The oscillator subassembly shall be subjected to the "Shock, transportation" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.2.

4.5.3 Temperature and humidity

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:01Z
Check the source to verify that this is the current version before use.

4.5.3.1 High temperature and humidity, nonoperating. The oscillator subassembly shall be subjected to the "High temperature and humidity, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.3.1.

4.5.3.2 High temperature, operating. The oscillator subassembly shall be subjected to the "High temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3,5.3.2.

4.5.3.3 Low temperature, nonoperating, The oscillator assembly shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.3.3.

4.5.3.4 Low temperature, operating, The oscillator subassembly shall be subjected to the "Low temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.5.3.4.

4.5.4 Immersion. The oscillator shall show no evidence of leakage while being subjected to the immersion test specified in MIL-STD-202, Method 104, condition A, and thereafter shall meet the performance and product requirements of this specification.

4.5.5 Operating life. The operating life of the oscillator subassembly shall be tested as follows:

4.5.5.1 Connect the oscillator subassembly as shown in Figure 1. Apply heater power to the vacuum tube and allow it to warm-up for at least 30 seconds. Apply 150 plus or minus 1 vdc as shown in Figure 1 and allow the circuit to operate for 30 plus or minus 1 minutes. Then remove the 150 vdc and allow the circuit to rest for at least 1 hour. This shall be considered one cycle of operation.

4.5.5.2 Repeat 4.5,5.1 until 800 cycles of operation have been completed, after which the oscillator subassembly shall meet all the requirements of 3.4.1 through 3.4.7.

4.5.5.3 The vacuum tube and test circuit components may be replaced as necessary to permit continuance of this test.

4.0 Waiver of tests. Tests specified in 4.5.1.4 and 4.5.2 shove may be waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.7 Functional tests. At the conclusion of the tests specified in 4.5.1.4 and 4.5.2 above, functional and other tests, as applicable, shall be conducted on the oscillator subassembly to determine freedom from operational malfunction and compliance with all performance requirements and product characteristics 01 this specification.

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:01Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging. The oscillator subassembly shall be packaged in accordance with Method III of MIL-P-116.

5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Unit packaging. The oscillator subassembly shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.2.1 Level A

5.2.1.1 Consolidation container, The oscillator subassemblies, packaged as decribed in 5.1.1.1, shall be packed in quantities as specified in the contract or order, in a wooden box in accordance with PPP-B-621, Class 2.

5.2.1.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Pattern No. 101 (MS 53102) (see 6.2).

5.2.2 Level C. The oscillator subassembly, packaged in accordance with 5.1.3.1, shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Levels A or C cushioning. The unit package shall incorporate sufficient cushioning material, bracing. or other adequate shock absorbing devices to insure that the oscillator subassembly will meet all the requirements of 4.5.1.4 (Vibration) and 4.5.2 (Shock). At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent.

5.4 Marking

5.4.1 Special markings. None, unless otherwise specified.

5.4.2 Normal markings. in addition to the markings required by me contract or order, unit packages and shipping containers shall be marked

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:01Z
Check the source to verify that this is the current version before use.

in accordance with the requirements Of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended use. 'The oscillator is intended for use in Torpedo Mark 37 Mods 0, 2, and 3.

6.2 Ordering date. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification.

(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing, including palletizing (see 5.2).

(e) Assigned activity for preproduction and periodic production inspections,

(f) Data required other than specified in 3.2.1.

additional samples required because of failure of any Unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

(h) Guarantee clause (see 6,4).

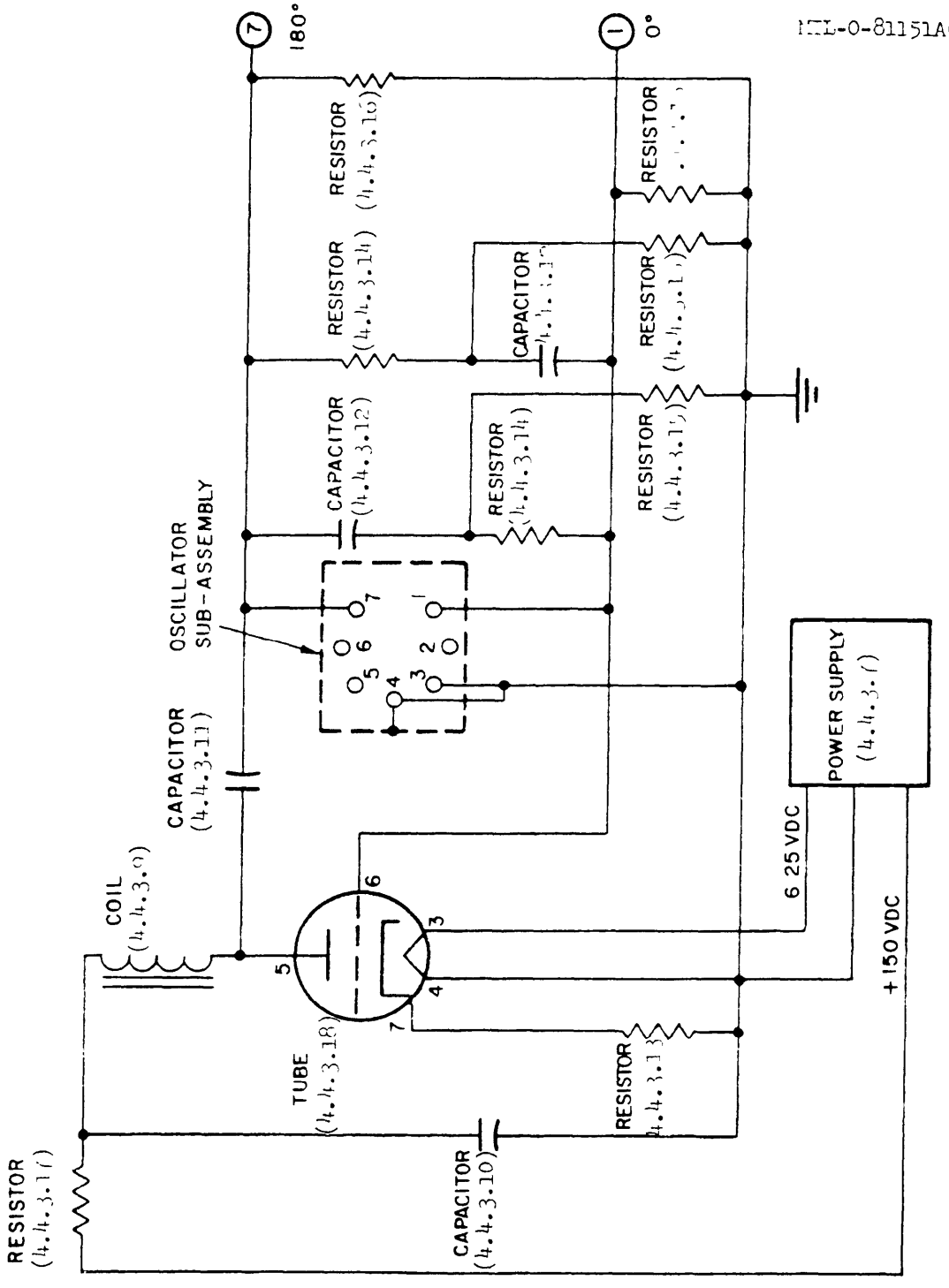
6.2,1 The attention of the contracting officer is invited to the options in 4.6.

6.3 Guarantee. The contract or order shall include a requirement for a guarantee stating that any oscillator which fails to meet the requirements of this specification within one year after delivery because of a defect in material or workmanship or because of the vendor's failure to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor et no cost to the Government.

6.4 The contract or order shall include a sufficient number of extra oscillators to cover preproduction and periodic production inspection samples.

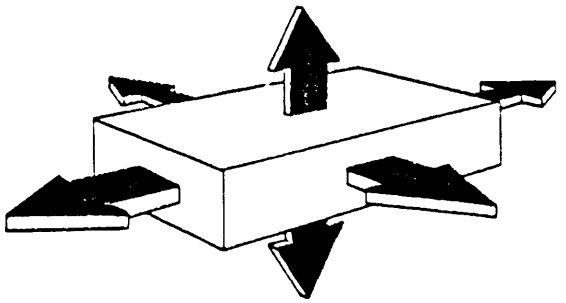
Preparing Activity
Navy - OS
Project No. 1355 - N169

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:01Z
Check the source to verify that this is the current version before use.

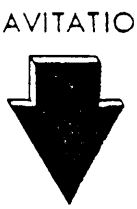


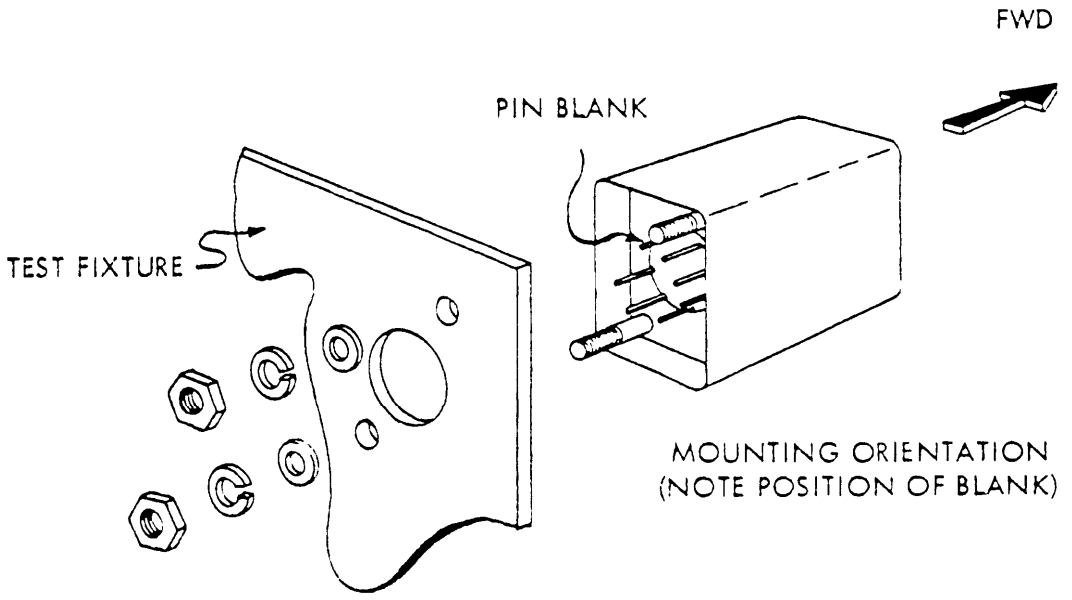
Source: https://assist.dla.mil -- Downloaded: 202640-23T05:01Z
Check the source to verify that this is the current version before use.

FIGURE 2. OSCILLATOR NAVORD DWG NO.1563736
MOUNTING REQUIREMENTS



DIRECTIONS OF VIBRATION INPUTS





NOTES:

1. FOR VIBRATION, ORIENTATION OF TEST ITEM WITH RESPECT TO EARTH'S GRAVITATIONAL FIELD SHALL BE MAINTAINED FOR ALL TEST DIRECTIONS.

2. FOR VIBRATION, THE TEST ITEM SHALL BE SECURED TO THE TEST FIXTURE AS SHOWN.

3. THE TEST FIXTURE SHALL NOT ALTER THE INPUTS FROM THE VIBRATION EXCITER TO THE MOUNTING POINTS OF THE TEST ITEM.

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:01Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:01Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:01Z
Check the source to verify that this is the current version before use.